Inventory-Constrained Dynamic Bidding for One-Shot Auctions

Rafi Putra Abdurrahman
AI and Data Engineering
Istanbul Technical University
abdurrahmanr23@itu.edu.tr
150230930

Emina Mandzukic
AI and Data Engineering
Istanbul Technical University
mandzukic23@itu.edu.tr
150230929

Emre Demirci
AI and Data Engineering
Istanbul Technical University
demirciem22@itu.edu.tr
150220317

I. Introduction

This project presents a conceptual solution to the problem of dynamic bidding in one-shot auctions [6] under an inventory constraints.

A one-shot auction is an auction in which each participant submits only one bid, and the highest bidder wins the product. Since each round is final, the company must carefully decide whether bidding is necessary and how much it should spend.

The products considered in this project are milk and dairy products, eggs, poultry, and beef. Since these goods are perishable, the company must maintain enough inventory while avoiding unnecessary excess stock.

The scenario considers an imaginary company that participates in daily auctions with other competitors within one country. The objective is not to win every auction, but to minimize total procurement cost while keeping the inventory of each product above a required minimum level.

Since competitors' inventory levels and auction quantities change over time, market prices are also dynamic. Therefore, the bidding rule must decide when to bid by considering both inventory needs and expected cost. Because competitors' bids are unknown, the model estimates them using least squares linear regression based on auction quantity and competitors' inventory levels.

The project aims to build a daily auction simulation, develop two bidding strategies with different levels of market information, satisfy inventory constraints, minimize procurement cost, and compare the strategies with a theoretical optimal solution.

II. Background

The Multi-Armed Bandit (MAB) framework [7] is a classic approach for making decisions over time when you do not know the outcomes in advance. The idea is to keep learning from past results and use that knowledge to make better choices in the future. It fits well in our setting because we also need to learn how the market behaves over time.

Lyapunov Optimization [8] is a technique for online optimization where you want to minimize cost over time while also keeping some system state, in our case, the inventory. Instead of solving the whole future at once, it provides a simple rule to follow at each step that is provably good in the long run.

III. Problem Formulation

The problem is broken down into three sub-problems, going from the simplified case to the most realistic one.

A. Deterministic Baseline (ILP)

In the first and simplest case, we assume that we already know everything: the quantities available at each auction and how much we would need to bid to win. This is not realistic, but it gives us the best possible solution to compare against. Since everything is fixed and known, the problem becomes a standard Integer Linear Programming (ILP) problem where we just decide at which rounds to bid. Let:

- $x_i^{(t)} \in \{0,1\}$ be the decision to bid for product i at time t,
- $b_i^{(t)}$ be the auction cost of product i at time t,
- $s_i^{(t)}$ be the inventory level of product i at time t,
- α_i be the minimum required inventory level,
- $q_i^{(t)}$ be the quantity sold at the auction,
- $d_i^{(t)}$ be the depletion of inventory.

$$\min_x \sum_{t=1}^{T} \sum_{i=1}^{N} b_i^{(t)} x_i^{(t)} \tag{1}$$

subject to $s_i^{(t)} \geq \alpha_i$ for all i, t and the dynamics $s_i^{(t+1)} = s_i^{(t)} + q_i^{(t)} x_i^{(t)} - d_i^{(t)}$. The solution to this acts as an offline optimal baseline.

B. All-Seeing Online Bidding

Here we move to the online setting, meaning we only see auction information one round at a time. We still assume that we can observe what the competitors are bidding at each round. To decide whether to bid, we use a form of Lyapunov Optimization [8]. We define a deficit state for each product that tracks how close we are to running out:

$$H_i(t) = \beta_i - s_i^{(t)}, \qquad \beta_i = \alpha_i + d_i^{\max} \tag{2}$$

Here β_i is set a bit above the hard minimum α_i to give us a safety buffer. This leads to the following bidding rule:

$$x_i^{(t)} = \begin{cases} 1, & \dfrac{b_i^{(t)}}{q_i^{(t)}} < \dfrac{H_i(t)}{V} \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

The idea is bid if the price per unit is low enough relative to how much we need the product. The hyperparameter $V > 0$ controls this balance, a higher V means we care more about saving money, a lower V means we prioritize keeping the inventory safe.

C. Partially Blind Online Bidding

In the most realistic case, we no longer see competitors' bids directly. We only know their inventory levels. Since we assume competitors bid based on how much stock they have, we use online ordinary least squares (OLS) regression to estimate what they would bid:

$$\hat{P}_i^{(t)} = \max_o \; K_o^\top \hat{\theta}_{i,o} \tag{4}$$

where K_o is the inventory of competitor o and $\hat{\theta}_{i,o}$ is the learned weight vector, updated after each round. We then bid $b_i^{(t)} = \hat{P}_i^{(t)} + \epsilon$ to just beat the estimated market price, and still use the same Lyapunov rule from All-Seeing Online Bidding to decide whether bidding is actually necessary.

IV. Datasets and Preprocessing

To create a realistic auction environment, this project uses real agricultural production data published by EUROSTAT, the official statistical office of the European Union. Four datasets are used, corresponding to beef, poultry, eggs, milk, and dairy products [1]–[4]. These datasets contain monthly production statistics for EU countries and provide a suitable basis for simulating auction-based mechanisms.

During preprocessing, the data quality flags were removed, all values were converted into numerical format, and values that could not be parsed were set to `NaN`. Missing values were handled using forward fill, while backward fill was used only when the missing value appeared at the beginning of the series.

Since the project assumes three auctions per day, monthly production values were converted into auction-level quantities. The monthly production was first divided by the number of days in the corresponding month, and then an auction factor of $1/20$ was applied. Each day was divided into three auction slots using a Dirichlet distribution [5], and Gaussian noise of approximately 5% was added to simulate market variation.

Because different products have different production volumes, all quantities were normalized to a standard scale of $[0, 1000]$. Since the products are perishable, the model also includes daily inventory losses. For Germany (DE), the approximate normalized depletion rates are 50.0 for milk, 25.0 for eggs, 48.0 for poultry, and 36.0 for beef.

If the EUROSTAT files are not available, the simulation uses synthetic data as a fallback option. In this case, α and $d_{\max}$ are defined using fixed synthetic parameters, and auction quantities are generated around predefined baseline values with approximately $\pm 20\%$ random noise.

V. Methodology

We first created the auction simulation in Python by utilizing data from the datasets described in IV. The problem was then separated into 3 subproblems with increasing complexity as described in III in which we solved from easiest to hardest sequentially.

A. Simulation Workflow

The simulation is implemented as a sequence of one-shot auctions over T rounds. At each time step t, the environment generates the auction quantities $q_i^{(t)}$ and depletion values $d_i^{(t)}$ for each product i, then stores the current inventories of all agents as the auction state. Competitor agents first generate their bids based on their own inventory conditions, after which our agent bids according to the information available in its settings: the All-Seeing agent observes competitor bids directly, while the Partially Blind agent only observes competitor inventories and estimates the market price. The auction is then resolved independently for each product, where the highest bidder wins and pays its own bid. Finally, every agent updates its inventory according to the inventory dynamic. The simulation records the cost, winning decisions, inventory levels, constraint violations, and any revealed information needed by the Partially Blind agent to update its online OLS estimator.

B. Opponent Behavior

Three opponent types are used in the simulation:

- **The stochastic competitor** bids inversely proportional to its inventory level and adds random noise to represent uncertain market behaviour.
- **The linear competitor** models its bid as a linear function of its inventory vector, meaning that low inventory usually leads to a higher bid.
- **The naive threshold competitor** follows a simpler rule: if its inventory is below the minimum threshold α_i, it submits a fixed high bid, otherwise, it does not bid seriously.

C. Deterministic Bidding with ILP

In the subproblem III-A's setting, we assume that the whole future is already known, including all auction quantities, depletion values, and the cost required to win each product at each round. Therefore, the problem becomes a deterministic Integer Linear Programming problem. For each product i and time step t, the binary decision variable $x_i^{(t)} \in \{0, 1\}$ determines whether the company buys the product or not. The objective is to minimize the total cost subject to the inventory safety constraint and the inventory transition. Since the products are treated independently and there is no shared budget constraint between them, the full optimization problem can be decomposed into N smaller single-product ILP problems. This makes the offline baseline easier to solve computationally. In the implementation, the ILP oracle solves these deterministic problems before the simulation starts, then replays the optimal schedule during the auction. Whenever $x_i^{(t)} = 1$, the oracle bids just above the known market price so that it wins the product.

D. All-Seeing Bandit-Style Problem

In order to solve the subproblem described in III-B, our approach is to utilize the Multi-Armed-Bandit (MAB) algorithm [7] while adapting a form of Lyapunov Optimization [8] so that it provides a way to optimize between cost minimization and inventory safety. The full formal analysis of our version is written in the Appendix.

For each product i, we define a deficit state,

$$H_i(t) = \beta_i - s_i^{(t)} \tag{5}$$

$$\beta_i > \alpha_i \tag{6}$$

where β_i is the soft inventory constraint to ensure the hard constraint α_i is always satisfied. The standard Lyapunov analysis becomes,

$$\min_{x_i^{(t)}} \sum_{i=1}^{N} \left(V b_i^{(t)} - H_i(t) q_i^{(t)} \right) x_i^{(t)} \tag{7}$$

Accordingly, the bidding rule becomes,

$$x_i^{(t)} = \begin{cases} 1, & \frac{b_i^{(t)}}{q_i^{(t)}} < \frac{H_i(t)}{V} \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

Here, V is a hyperparameter that controls the importance between cost minimization and inventory safety. If V is high, then more emphasis on cost minimization is given, and vice versa. This bidding rule will prioritize buying product i if it is considerably cheap while maintaining the hard inventory constraint.

E. Partially Blind Bandit-Style Problem

For the last subproblem described in III-C, we can assume that the other buyers bid linearly w.r.t. to their inventory conditions (and we will model the environment as such). Thus, a trivial solution is to approximate the market price (highest bid among other buyers) for an auction selling product i as,

$$\mathbb{E}[C_{i,o} | K_o] = K_o^T \theta_{i,o} \tag{9}$$

where,

- C is the cost distribution for buyer o and product i.
- K is the context distribution, in this case the buyer o's inventory condition.
- θ is the learnable parameter vector for pair product i and buyer o.

The market price for product i at time t is then

$$P_i^{(t)} = \max_o \mathbb{E}[C_{i,o} | K_o] \tag{10}$$

and our bid becomes

$$b_i^{(t)} = P_i^{(t)} + \epsilon \tag{11}$$

where ϵ is the smallest number that guarantees win. Here, we can get data and fit this linear regression model online using incremental Ordinary Least Squares (OLS).

TABLE I
RUNTIME COMPARISON OF THE FOUR BIDDING STRATEGIES

Algorithm	Total Cost	Violations
ILP Oracle	1,641.30	0
Static Plan ILP	1,630.19	32
All-Seeing Bandit	3,136.68	0
Partially Blind Bandit	252,216.72	0

VI. RESULTS

A. Runtime Results

The runtime comparison is conducted by evaluating the ILP oracle, static-plan ILP, All-Seeing Bandit, and Partially Blind Bandit using the following settings:

- Auction rounds: T = 150
- Lyapunov Hyperparameter: V = 2.0

Examining Table I, the ILP oracle achieves a total cost of $1,641.30$ with zero violations, making it the best feasible baseline. Although the static-plan ILP obtains a slightly lower cost of $1,630.19$, it produces 32 violations, meaning that the lower cost is obtained by failing to satisfy the inventory constraint. The All-Seeing Bandit achieves zero violations with a total cost of $3,136.68$, which is higher than the ILP oracle but still reasonable for an online method that does not know the future. Since it observes competitor bids directly, it can use the Lyapunov threshold rule with accurate market prices. The Partially Blind Bandit also avoids violations, but its cost increases sharply to $252,216.72$. This shows that estimating competitor bids only from inventory information is much harder. In this case, the agent becomes overly conservative and bids too aggressively to avoid stockouts, leading to a very large procurement cost.

B. Lyapunov Hyperparameter Analysis

The effect of the Lyapunov hyperparameter V is analyzed over 500 auction rounds. The results are shown in Table II.

TABLE II
EFFECT OF V ON THE ALL-SEEING AND PARTIALLY BLIND AGENTS OVER 500 ROUNDS

V	AS Cost	PB Cost	Violations
0.1	10,404.16	4,745,758.38	0
2.0	10,400.98	271,348.39	0
5.0	10,735.06	130,128.17	0
10.0	10,735.06	82,377.65	0

As shown in the table, all tested values of V satisfy the inventory constraints with zero violations. However, the cost behaviour is different for the two agents. The All-Seeing agent remains relatively stable, with total cost staying around $10,400$–$10,735$. This is because it observes the exact competitor bids and can make accurate bidding decisions even when V changes. In contrast, the Partially Blind agent is much more sensitive to V. When $V = 0.1$, its total cost becomes extremely high because the agent prioritizes inventory safety too strongly and overbids based on uncertain market estimates.

As V increases, the Partially Blind agent becomes more cost-conscious, reducing its cost from $4,745,758.38$ to $82,377.65$ while still keeping zero violations. This suggests that larger V values are more suitable for the Partially Blind setting, since they reduce excessive bidding caused by inaccurate price estimation.

C. Convergence Analysis

The convergence behavior of the online agents is analyzed using the average regret per round, $\mathrm{Reg}(T)/T$, over 500 auction rounds. The regret measures how much worse an online bidding strategy performs compared to the offline ILP oracle. Since the ILP oracle has complete future information, it is used as the optimal reference. Therefore, the cumulative regret can be written as

$$\mathrm{Reg}(T) = C_{\mathrm{agent}}(T) - C_{\mathrm{ILP}}(T), \tag{12}$$

where $C_{\mathrm{agent}}(T)$ is the total cost paid by the online agent up to round T, and $C_{\mathrm{ILP}}(T)$ is the total cost paid by the offline ILP oracle over the same horizon. A smaller average regret means that the online agent is getting closer to the oracle performance. In the no-regret setting, the desired behavior is for the average regret to behave as

$$\lim_{T\to\infty} \frac{\mathrm{Reg}(T)}{T} = 0, \tag{13}$$

meaning that the cost difference per round becomes negligible as the number of auction rounds increases.

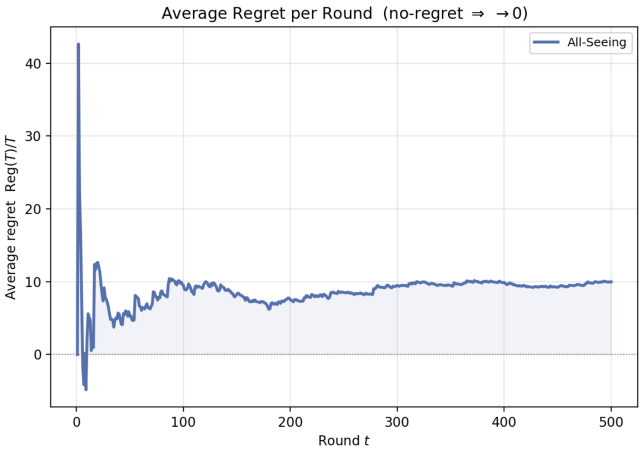


Fig. 1. Average regret per round of the All-Seeing agent over 500 rounds with V set to 2.0.

For the All-Seeing agent, the average regret initially fluctuates because early auction decisions have a large effect when T is still small. After this early period, the curve becomes more stable and stays around a small positive value. This behavior is consistent with the Lyapunov regret bound given in the Appendix.

The Partially Blind agent shows a different pattern. Its average regret is very high at the beginning, but decreases sharply as the number of rounds increases. This happens because the agent initially has very limited information for

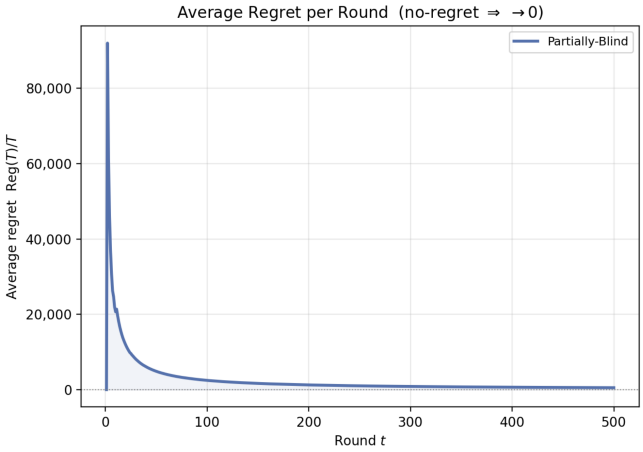


Fig. 2. Average regret per round of the Partially Blind agent over 500 rounds with V set to 2.0.

estimating competitor bids. As more winning bids are revealed, the incremental OLS estimator receives more samples and the market-price prediction becomes more accurate. This behavior follows the expected regret structure given in the Appendix.

where the second term represents the additional estimation penalty caused by hidden competitor bids. As n increases, this penalty becomes smaller, explaining the decreasing trend in Fig. 2. Thus, although the Partially Blind agent has much higher early regret than the All-Seeing agent, the curve shows that the online learning mechanism gradually improves its bidding decisions.

VII. Discussion

Overall, the project works as intended and shows that you can make decent bidding decisions using only inventory information and a simple rule. On top of that, there are a few things that could be improved.

Competitor modeling. Right now we assume competitors bid linearly based on their inventory. This is a simplification. In reality their behavior could be more complex, and using something like a neural network to estimate their bids might give better results.

Multi-item bidding. Each product is handled completely separately. In practice, a company's need for one product might affect how it bids on another. Handling all products together in one joint decision could make the strategy better.

Budget constraints. In our current setup, the company has no limit on how much it can spend in total. In a more realistic scenario, there would be a maximum budget that cannot be exceeded. Adding this constraint would change the problem a lot, since the company would also need to decide how to spread its spending over time, not just whether to bid at a given round.

Real data. The auction quantities and prices in our simulation are derived from production statistics and synthetic noise. It would be interesting to test the same approach on real auction data to see how well it actually performs.

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APPENDIX

Optimality Analysis

We want to satisfy the hard constraints $s_i^{(t)} > \alpha_i \; \forall_{t,i}$ while minimizing total cost over time. Thus, we need to define a state function $H(t)$ that gives the difference of current s and its respective α. Given the inventory dynamics

$$s_i(t+1) = s_i(t) + q_i(t)x_i(t) - d_i(t),$$

we can define the deficit state function w.r.t. hyperparameter β_i (an analysis of this parameter will be provided shortly)

$$H_i(t) = \beta_i - s_i(t),$$

$$\beta_i > \alpha_i.$$

Then, by definition,

$$H_i(t+1) = H_i(t) - q_i(t)x_i(t) + d_i(t).$$

By this we can define the Lyapunov function

$$L(H(t)) = \frac{1}{2}\sum_{i=1}^{N} H_i(t)^2,$$

Since our current decision $x_i(t)$ depends on $H_i(t)$, we can define the one-step conditional drift

$$\Delta(H(t)) = \mathbb{E}[L(H(t+1)) - L(H(t)) \mid H(t)].$$

In order to satisfy the constraint, an equivalent goal is to minimize $\Delta(H(t))$ with β big enough that the constraint is always satisfied.

To achieve the main goal of minimizing cost over time, we can define a new target

$$J(x(t)) = \mathbb{E}\left[\sum_{i=1}^{N} b_i(t)x_i(t) \mid H(t)\right]$$

Using these targets, we can create a new weighted target. We can assume $0 \le q_i(t) \le q_i^{\max}$ and $0 \le d_i(t) \le d_i^{\max}$ for all i, t. Then there exists a finite constant $B > 0$ such that

$$\Delta(H(t)) + V\,J(x(t))$$
$$\le B + \sum_{i=1}^{N} H_i(t)\mathbb{E}[d_i(t) \mid H(t)]$$
$$+\mathbb{E}\left[\sum_{i=1}^{N}\big(Vb_i(t) - H_i(t)q_i(t)\big)x_i(t)\middle| H(t)\right].$$

Here $V > 0$ is the hyperparameter that determines the importance of minimizing $J(x(t))$ over satisfying the constraints, represented with $\Delta(H(t))$. We will provide the derivation shortly. Since only the last part of the right side of the inequality is under our control, minimizing it is equivalent to minimizing the left side

$$\min[\Delta(H(t)) + V\,J(x(t))] \equiv \min_{x(t)} \sum_{i=1}^{N}\big(Vb_i(t) - H_i(t)q_i(t)\big)x_i(t).$$

If $x_i(t) \in \{0, 1\}$ and the decisions are separable across i, then the optimal policy is

$$x_i(t) = \begin{cases} 1, & H_i(t)q_i(t) > Vb_i(t), \\ 0, & H_i(t)q_i(t) \le Vb_i(t). \end{cases}$$

Equivalently, for $q_i(t) > 0$,

$$x_i(t) = 1 \quad \Longleftrightarrow \quad \frac{b_i(t)}{q_i(t)} < \frac{H_i(t)}{V}.$$

Here, the bidding rule is to bid whenever the price of product is cheap and/or the deficit state is high.

Proof of Upper Bound

From

$$H_i(t+1) = H_i(t) - q_i(t)x_i(t) + d_i(t),$$

we have

$$(H_i(t+1))^2 = (H_i(t))^2 + \big(d_i(t) - q_i(t)x_i(t)\big)^2$$
$$+2H_i(t)\big(d_i(t) - q_i(t)x_i(t)\big),$$

subtracting both sides by $(H_i(t))^2$ gives

$$(H_i(t+1))^2 - (H_i(t))^2 = \big(d_i(t) - q_i(t)x_i(t)\big)^2$$
$$+2H_i(t)\big(d_i(t) - q_i(t)x_i(t)\big).$$

Summing over i, multiplying by $1/2$, and taking conditional expectation yields

$$\Delta(H(t)) = \mathbb{E}\left[\frac{1}{2}\sum_{i=1}^{N}\big(d_i(t) - q_i(t)x_i(t)\big)^2\middle| H(t)\right]$$
$$+\sum_{i=1}^{N} H_i(t)\mathbb{E}[d_i(t) - q_i(t)x_i(t) \mid H(t)].$$

By the assumption of bounds of $d_i(t)$ and $q_i(t)$, there exists $B < \infty$ such that

$$\mathbb{E}\left[\frac{1}{2}\sum_{i=1}^{N}\big(d_i(t) - q_i(t)x_i(t)\big)^2 \middle| H(t)\right] \le B.$$

Hence

$$\Delta(H(t)) \le B + \sum_{i=1}^{N} H_i(t)\mathbb{E}[d_i(t) \mid H(t)]$$

$$-\mathbb{E}\left[\sum_{i=1}^{N} H_i(t)q_i(t)x_i(t) \mid H(t)\right].$$

Adding $V\,J(x(t))$ to both sides gives

$$\Delta(H(t)) + V\,\mathbb{E}\left[\sum_{i=1}^{N} b_i(t)x_i(t) \mid H(t)\right]$$

$$\le B + \sum_{i=1}^{N} H_i(t)\mathbb{E}[d_i(t) \mid H(t)]$$

$$+\mathbb{E}\left[\sum_{i=1}^{N}\big(Vb_i(t) - H_i(t)q_i(t)\big)x_i(t) \middle| H(t)\right].$$

The first two terms on the right-hand side are independent of the control action, so minimizing the drift-plus-penalty upper bound reduces to minimizing

$$\sum_{i=1}^{N}\big(Vb_i(t) - H_i(t)q_i(t)\big)x_i(t).$$

If decisions are separable and binary, the minimizer is obtained by choosing $x_i(t) = 1$ whenever

$$Vb_i(t) - H_i(t)q_i(t) < 0,$$

which is equivalent to

$$H_i(t)q_i(t) > Vb_i(t).$$

Analysis of β

As one can see, minimizing $\Delta(H(t))$ is equivalent to minimizing $L(H(t+1)$. L here is quadratic, meaning that minimizing this target can be seen geometrically as bounding q_i to always be in some radius around β_i. Thus, if $\beta_i - \alpha_i$ is not big enough, the constraint can be violated.
Since we know that the radius mentioned is determined by the supply s_i's dynamic, a possible solution to this is to set

$$\beta_i = \alpha_i + d_i^{\max}$$

since d_i is assumed to be bounded, the supply dynamic is always $q_i(t)x_i(t) - d_i(t)$, and the lower bound over all time t of the dynamic is $-d_i^{max}$.
A question may arise: "Why not buy every time the price is cheap?". Here, as in real-life, food has expiration and is included in d_i, so it is not a wise decision to overstock on food.

Regret Bounds

Let:

- T: total number of auction rounds in the simulation.
- V: Lyapunov hyperparameter.
- N: number of products considered in the auction problem.
- M: number of competitors in the auction environment.
- $K_{\max}$: maximum size or norm of the observed context vector used in the Partially Blind setting.
- d: dimension of the context vector used by the online OLS estimator.
- n: number of observed samples used to estimate competitor bids.

1) All-Seeing Bandit:

$$\mathbb{E}\left[\mathrm{Reg}_T^{\mathrm{Lyap}}\right] = O\left(\frac{T}{V} + V\right).$$

2) All-Seeing Bandit:

$$\mathbb{E}\left[\mathrm{Reg}_T\right] = O\left(\frac{T}{V} + V\right) + O_p\left(TNK_{\max}\sqrt{\frac{d + \log(NM)}{n}}\right)$$